

The Hidden Trading Activity Puzzle: Bond ETFs versus Equity ETFs^{*}

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Abstract

We examine the trading dynamics in exchange-traded funds (ETFs) with regards to hidden trading activity. Specifically, we compare the level of hidden activity in bond ETFs and equity ETFs. We find the percentage of hidden volume in bond ETFs to be significantly higher than in equity ETFs (41% versus 28%, respectively). Using high frequency data, we find that trading costs (quoted, effective, realized spread, and price impact) and pinging activity are lower for bond ETFs than equity ETFs, yet bond ETFs are traded more opaquely. Results are robust to out-of-sample tests and are not driven by a specific exchange or specific types of bond ETFs.

JEL Classifications: G10, G12, G14

Keywords: Hidden Activity; Dark Trading; Exchange-Traded Funds; Trading Costs; Algorithmic Trading

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1. Introduction

Hidden trading accounts for nearly 40% of trading volume (Bartlett and O’Hara, 2024), and exchange-traded funds (ETFs) represent over 10% of the market capitalization of securities traded on U.S. exchanges, accounting for more than 30% of the overall daily trading volume.¹ This paper examines the trading activity of the two largest types of U.S.-listed ETFs: bond ETFs and equity ETFs.² ETFs, like equities, trade on traditional lit exchanges, where bid and ask quotes are publicly posted. Additionally, traders of ETFs have the option to place hidden orders on lit exchanges or trade in off-exchange execution venues, such as dark pools, where price quotes are not publicly disclosed (Bloomfield, O’Hara, and Saar, 2015; and Lee and Chung, 2022).³

We focus on bond and equity ETFs, as there are informational differences in bond and equity ETFs which may result in differences in trading dynamics. Downing, Underwood, and Xing (2009) show that the corporate bond market is less informationally efficient than the equity market. While both bond and equity ETFs trade on exchanges, their underlying assets trade in different venues. Bonds trade over-the-counter with less transparency than equities, while equities trade on exchanges with more transparency than bonds. The purpose of this paper is to see if there are differences in trading between bond and equity ETFs, and if so, explain the differences.

We find that bond ETFs have a higher percentage of hidden activity (as proxied by the percentage of hidden volume) than a matched set of equity ETFs. We examine differences in bond and equity ETFs to explain the puzzle as to why bond ETFs would have a higher

¹ See <https://www.nasdaq.com/articles/global-etf-market-facts%3A-three-things-to-know-from-q1-2023>

² The Assets Under Management (AUM) Leaderboard of ETFs on <https://etfdb.com/etfs/asset-class/> show that AUM for equity (bond) ETFs is \$6.7 trillion (\$1.5 trillion).

³ Hidden orders follow price, display, and time precedence rules (Lee and Chung, 2022).

percentage of hidden volume (*Hidden Volume (%)*) than equity ETFs and are unable to explain this puzzle. Bond ETFs and equity ETFs have underlying assets that trade in different markets, although the ETFs trade in the same market; bond and equity ETFs have relatively consistent differences in *Hidden Volume (%)*.

Bonds in the U.S. trade primarily in the over-the-counter (OTC) bond market. This market is characterized as opaque and illiquid with high transaction costs (Bao, Pan, and Wang, 2011; Bessembinder, Spatt, and Venkataraman, 2020; and Edwards, Harris, and Piwowar, 2007). The introduction of the Trade Reporting and Compliance Engine (TRACE) in 2002 made bond transactions data publicly available, establishing post-trade transparency in the OTC bond market. Nevertheless, the bond market does not provide pre-trade transparency and lags behind the equity market in terms of informational efficiency as bond prices adjust slower than equity prices (Downing, Underwood, and Xing, 2009). The dominance of institutional investors, who trade in the OTC bond market, has limited the adoption of electronic trading in bonds (O'Hara and Zhou, 2021; Biais and Green, 2007; and Hendershott and Madhavan, 2015).

Market participants on stock exchanges have embraced reduced transparency both on- and off-exchanges through hidden orders on exchanges and hiding orders in dark pools (Degryse, Karagiannis, Tombeur, and Wuyts, 2021). Traders may opt to submit hidden orders or trade in dark pools for several reasons. First, submitting hidden orders or trading in dark pools can help traders avoid private information leakage, an increase in trading costs, and a decrease in liquidity due to front-running from opportunistic traders (Boulatov and George, 2013; and Harris, 1996). In addition, traders may use dark pools, which have limited access, to reduce the risk of trading against informed order flow while retaining the ability to trade inside the lit market spread (Buti, Rindi, and Werner, 2022). Cox (2022) finds that hidden volume executed on the NYSE Chicago

(CHX) represents 11.07% and 2.38% of all hidden volume across lit exchanges for ETFs and stocks, respectively. Cox concludes that the high level of hidden activity for ETFs on CHX is a result of cross trades in multi-leg option strategies. Therefore, his findings imply that sophisticated investors use hidden orders to trade ETFs.

Given the different market structures for the underlying securities in bond and equity ETFs, we anticipate differences in the trading of these ETFs, even though both types of ETFs trade in the equity market. Traders have the option to trade bond ETFs transparently on lit exchanges or opaquely via hidden orders or dark pools. There may be many factors that influence order choice, such as market conditions, price aggressiveness, or order size (Bessembinder, Panayides, and Venkataraman, 2009; and De Winne and D'Hondt, 2007).

Investor types also play a role in this choice as institutional investors have the option to submit hidden orders while retail investors do not. Gempesaw, Henry, and Xiao (2024) find that the average retail dollar volume weight in equity ETFs is 28.6% relative to 11.7% for fixed income ETFs which shows that retail traders are attracted to equity ETFs more than bond ETFs.

We investigate the aggregate level of hidden trading activity in bond and equity ETFs and explore differences in hidden activity between the two types of ETFs. Only orders that execute fully or partially against hidden orders are identified.⁴ We find that, on average, bond ETFs have significantly more *Hidden Volume (%)* than equity ETFs (41% versus 28%, respectively). Given the lack of pre-trade transparency in the OTC bond market, bond traders are accustomed to trading bonds opaquely, which could affect how bond ETFs trade. Although spread, trade size,

⁴ MIDAS reports only the executed portion of hidden orders. Data limitations do not allow us to analyze all nondisplayed orders. We focus on the portion of hidden orders that successfully transact. This limitation also prevents us from looking at other aspects of the order exposure decision, such as price aggressiveness and order size. We refrain from using hidden trades as this may be an inflated measure (e.g. there may be many small trades that transact against one large hidden order) but untabulated results using percentage hidden trades to total trades (*Hidden Trades (%)*) as an alternative hidden liquidity proxy remain consistent with the main sample analysis.

assets under management (AUM), and ETF age can explain some of the differences in *Hidden Volume (%)* in the ETFs, traders appear to have a preference for trading bond ETFs opaquely.

Intraday analyses reveal that bond ETFs have lower trading costs than equity ETFs, and the differences in trading costs explains some of the higher *Hidden Volume (%)* in bond ETFs. We find evidence that pinging activity is negatively associated with *Hidden Volume (%)* in bond and equity ETFs, indicating that potential algorithmic trading (AT) is not increasing the percentage of hidden activity.

We also conduct robustness tests to ensure that our finding of higher *Hidden Volume (%)* in bond ETFs is not driven by other factors. We perform two out-of-sample tests, one of which is in 2012 and the other from 2019 to 2020. Hidden activity is higher in bond ETFs than equity ETFs in both out-of-sample periods. We compare *Hidden Volume (%)* between the two types of ETFs across the lit exchanges and find that the levels of hidden activity of bond and equity ETFs are not significant between exchanges. This result implies that traders do not have a preference for trading a particular type of ETF on a particular exchange. We separate our bond ETF sample into “more risky” corporate bond ETFs and “less risky” government bond ETFs and compare each type to a matched sample of equity ETFs. We find some evidence that more risky corporate bond ETFs trade more opaquely than less risky government bond ETFs and that both types of bond ETFs trade more opaquely than equity ETFs.

Our research is related to several strands of literature. First, a number of studies relate the use of hidden orders (activity) to price transparency and information asymmetry. Boulatov and George (2013) posit that while hidden orders may seem to benefit informed traders who do not want to expose their information, allowing for hidden orders facilitates liquidity provision by the informed, which improves market quality through competition. Hendershott and Jones (2005)

study Island electronic communications network (ECN) when it stopped displaying its limit order book for the three most active ETFs in September 2002. Hendershott and Jones find that ETF prices adjust more slowly, trading activity shifts to other venues, and trading costs increase. Island ECN's decision to go "dark" shows that a reduction in transparency can reduce the price discovery process. While we expect bond ETFs to benefit from the increased transparency of the lit exchanges when compared to the OTC bond market, we find that bond ETFs trade more opaquely than equity ETFs. Our findings support the notion that assets that are more difficult to value will trade more opaquely.

Second, our analyses also relate to the empirical and theoretical literature on market fragmentation. U.S. markets are increasingly fragmented with multiple lit exchanges and dark trading venues. Degryse, de Jong, and van Kervel (2015) argue that fragmentation in lit markets improves liquidity through competition from liquidity suppliers, while dark trading negatively affects liquidity. The model of Buti, Rindi, and Werner (2017) show that dark pool activity increases in the dispersion of traders' valuations of an asset. We conduct analyses on bond and equity ETFs' trading activity across lit exchanges and off-exchange dark pools. We find that bond ETFs trade more in dark pools than equity ETFs. More dark pool trading of bond ETFs may indicate a substitution effect for hidden orders on exchanges or a complementary effect that provides indirect evidence of trader preferences to trade bond ETFs opaquely.

Third, our study has regulatory implications in that the prevalent use of hidden orders and dark pool trading has raised concerns of market instability and reduced informational efficiency (Bloomfield, O'Hara, and Saar, 2015). Being able to trade bonds via ETFs expands the reach of the bond market from that of the OTC market, dominated by institutional investors, to that of lit exchanges. The ease of trading ETFs raises concerns about the unintended consequences for the

underlying securities in ETF baskets (Dannhauser, 2017). The increased liquidity that ETFs offer may lead to demand shocks in the underlying securities and cause price volatility (Ben-David, Franzoni, and Moussawi, 2018). This volatility may materialize in ETF prices and cause shifts in trading behavior (e.g., an increase in the use of hidden orders, Degryse, Karagiannis, Tombeur, and Wuyts, 2021). Hidden trading and dark pools are the purview of institutional investors, which leads to the debate of an uneven playing field among market participants.

The remainder of the paper proceeds as follows. Section 2 provides a detailed review of the literature on ETFs and trading features on exchanges and outline the development of the hypotheses. Section 3 describes the data and sample selection process. Section 4 outlines the methods used to conduct the analyses. Section 5 presents the main empirical findings. Section 6 details the robustness tests and section 7 concludes.

2. Literature Review and Hypotheses Development

Research on ETFs primarily focuses on equity ETFs. Ben-David, Franzoni, and Moussawi (2018) find that equities with higher ETF ownership have higher volatility. Huang, O'Hara, and Zhong (2021) conclude that equity industry ETFs can help hedge risk and improve market efficiency. Box, Davis, Evans, and Lynch (2021) find little evidence that equity ETFs impact returns on the underlying equities. Ahn and Patatoukas (2022) find that stock indexing leads to price discovery in micro-cap stocks. Two studies that look at short selling activity in equity ETFs (Karmaziene and Sokolovshi, 2022; and Li and Zhu, 2022) show that equity ETFs can be used to circumvent short-sale constraints of underlying stocks. Converse, Levy-Yeyati, and Williams (2023) are one of the few studies that include bond and equity ETFs (2023) in their study, which compares ETFs to mutual funds.

ETFs, including both bond ETFs and equity ETFs, trade on exchanges that provide pre-trade transparency, high liquidity, and low transaction costs. Due to the structure of the OTC bond market being different from modern equity exchanges, bond ETFs are a financial innovation that bridges the bond market to the equity market. The popularity of bond ETFs relative to the underlying bonds shows bond ETFs as an alternative to investing in bonds directly (Dannhauser, 2017).⁵

The introduction of TRACE brought about a reduction in trading costs for bonds with their transactions disseminated through the reporting system (Edwards, Harris, and Piwowar, 2007; and Goldstein, Hotchkiss, and Sirri, 2007), showing that transparency is beneficial to traders. Nevertheless, as the bond market becomes more transparent, the equity market is embracing opacity (Bessembinder, Spatt, and Venkataraman, 2020; and Buti, Rindi, and Werner, 2022). The use of hidden orders is pervasive on equity exchanges and trading in dark pools has gained momentum (Jain and Jain, 2014; and Rosenblatt Securities, 2023). Using a sample of Euronext-Paris stocks, Bessembinder, Panayides, and Venkataraman (2009) show that hidden orders have value to traders as hidden orders represent 44% of their sample order volume. On U.S. stock exchanges, 13% of trades execute against hidden orders and 14% of order volume is hidden (Jain and Jain, 2017). More recently, Bartlett and O'Hara (2024) find that hidden orders provide liquidity for approximately 40% of trading volume. In the U.S., off-exchange market share in 2023 increased to 43.97%.⁶ The pervasive and extensive use of hidden orders and dark pools show that traders value the ability to hide orders.

Bond ETFs provide traders the opportunity to transparently trade bonds via bond ETFs. Holden and Nam (2020) find that the introduction of bond ETFs increases the liquidity of the

⁵ See “4 Trends Driving ETF Growth” on <https://www.blackrock.com/hk/en/ishares/insights/growth-trends>.

⁶ See <https://www.rblt.com/market-structure-reports/let-there-be-light-us-edition-54>

underlying bond market with positive liquidity spillover effects being greatest for low volume, high yield, and 144A bonds. If traders value the transparency that bond ETFs provide, then they will be less likely to use hidden orders when trading bond ETFs.

We believe hidden activity in bond ETFs will be low if bond traders value the transparency offered by bond ETFs and consequently trade bond ETFs rather than individual bonds. For this reason, we expect hidden activity in bond ETFs to be lower than in equity ETFs.

H1: Hidden activity, as measured by the percentage of hidden volume on lit exchanges, is higher for equity ETFs than bond ETFs.

The use of dark trading venues and hidden trading on exchanges can be considered substitutes (Degryse, Karagiannis, Tombeur, and Wuyts, 2021). Kwan, Masulis, and McInish (2015) find the opportunity to bypass limit order queues on traditional exchanges is one of the reasons for the rapid rise of dark pools. If trading in dark pools is a substitute for hidden trading on exchanges, we expect equity ETFs to have more dark trading than bond ETFs.

H2: Dark trading in equity ETFs is higher than in bond ETFs.

If bond or equity ETFs have more hidden activity than the other, then we expect other trading differences to exist. Lee and Chung (2022) find that the use of hidden liquidity is associated with an increase in spreads (quoted, effective, and realized), informed trading, trading volume and trade size. Hidden activity on exchanges increases with stock volatility, which should also apply to ETFs (Degryse, Karagiannis, Tombeur, and Wuyts, 2021; and Menkveld, Yueshen, and Zhu, 2017). If hidden activity is different in bond ETFs and equity ETFs, then we posit that trading metrics, including trade volume, the number of trades, spread, and volatility, will also be different. ETF characteristics such as expense ratio, AUM, and fund age may affect trading as ETFs with lower expense ratios, higher AUM, and longer history may be more

popular due to more information available on performance histories and be more salient to traders. Differences in ETF characteristics may reveal differences in hidden activity between bond and equity ETFs.

H3: Trading metrics and fund characteristics may be related to the differences in hidden activity between bond ETFs and equity ETFs.

The Securities and Exchange Commission (SEC) documents an increase in hidden volume from 15% to over 50% between 2012 to 2021 (Chakrabarty, Hendershott, Nawn, and Pascual, 2025). Traders may choose to hide their trading intentions for several reasons: to avoid information leakage, pick-off risk, and price impact (Harris, 1996). We examine whether differences in trading costs drive different levels of hidden activity (*Hidden Volume (%)*) in bond and equity ETFs. Traders may use hidden orders to reduce the price impact of their ETF trades. If traders of ETFs prioritize reducing the price impact of their trades, then we would observe higher levels of *Hidden Volume (%)* executions.

H4: Trading costs may be related to differences in hidden activity between bond ETFs and equity ETFs.

Algorithmic trading (AT) extracts information from other market participants and benefit from short-term price movements (Weller, 2018). Boehmer, Li, and Saar (2018) use the return and volatility of an index ETF to identify HFT firms engaging in short-term directional speculation strategies. High levels of hidden orders may attract trading algorithms to engage in pinging activity as sophisticated software programs seek to identify profitable trading opportunities (Chakrabarty, Hendershott, Nawn, and Pascual, 2025; Lee and Chung, 2022). Using the four Weller (2018) algorithmic trading proxies and potential pinging activity, we examine the level of algorithmic trading in the bond and equity ETFs.

H5: High levels of hidden trading activity in ETFs are correlated with algorithmic trading.

3. Data and Sample

3.1 Data Sources and Sample

We obtain trading data on lit exchanges from the SEC's Market Information Data Analytics System (MIDAS) for the sample period from January 2021 to March 2022. Trade volume is identified in MIDAS along with security characteristics such as ticker symbols, trading dates, and security identifier for stock or ETF). Dark trading data consisting of ATS Tier 1 securities is obtained from FINRA. In the intraday analysis, high-frequency data is from NYSE's Daily Trade and Quote (TAQ) database for the sample period from January 2022 to March 2022. The TAQ data is filtered using only trades and quotes during the times the market is open (9:30 am until 4:00 pm).

Information on ETF prices, number of shares outstanding, total volume, ask price, bid price, high price, and low price are obtained from the CRSP daily security files. To distinguish bond ETFs and equity ETFs from other types of ETFs, we use the asset class list compiled on the ETF database website.⁷ ETF characteristics, such as fund inception year, AUM, and expense ratio, are from the ETF database website.

ETFs have to be priced at or above \$5 and trade every day in the sample period to be included in the sample. There are 314 trading days in the main analysis and 62 trading days in the intraday analysis. There are a total of 1,182 unique bond and equity ETFs that satisfy the filter conditions of which 962 are equity ETFs and 220 are bond ETFs.

⁷ See <https://etfdb.com/etfs/asset-class/>.

To ensure that certain characteristics are not driving the results, we create a matched sample of bond and equity ETFs based on price and market capitalization.⁸ Davies and Kim (2009) find that the best practice is to match firms one-to-one (without replacement) based on share price and market capitalization.⁹ To obtain a matched sample based on price and market capitalization, we calculate the following score from Huang and Stoll (1996) for each bond ETF using the entire sample of equity ETFs,

$$score = \sum_{i=1}^2 \left[\frac{(Y_i^B - Y_i^E)}{(Y_i^B + Y_i^E)/2} \right]^2, \quad (1)$$

where Y_i represents ETF attributes, B and E refer to bond ETFs and equity ETFs, respectively, and Σ denotes the summation over i attributes. Then, for each bond ETF, we select an equity ETF with the smallest score as its match (without replacement). The main matched sample consists of 194 matched pairs, which is a total of 388 ETFs. The intraday sample consists of 293 matched pairs or 586 total ETFs.¹⁰ Table 1 reports the types and grade classifications of the sample's 194 bond ETFs. The classification information is found on the ETF Database. Bond type is found under FactSet Classifications for Category and bond grade is found under FactSet Classifications for Focus. The most frequent bond ETF type in our sample is corporate, with 74 corporate bond ETFs, and the next most frequent type of bond ETF is broad, with 60 broad bond ETFs. Most bond ETFs in the sample are investment grade (130 bond ETFs).

3.2 Variable Construction

⁸ In untabulated analyses, we match on the two variables, price and market capitalization, separately. The matched samples are constructed based on the smallest difference in the matching variables. Both matched samples have 220 matched pairs. Our results hold across the different matching specifications.

⁹ Bartlett and O'Hara (2024) show that hidden activity is related to share price.

¹⁰ In untabulated intraday analysis, we create a second matched sample based on the smallest difference in market capitalization, which also has 293 matched pairs.

Hidden activity is proxied by *Hidden Volume (%)*, hidden volume divided by total trade volume.¹¹ The four algorithmic trading proxies, the odd-lot volume ratio, the trade-to-order volume ratio, the cancel-to-trade ratio, and the average trade size, are constructed following Weller (2018). According to Weller, high odd-lot volume ratios and cancel-to-trade ratios indicate high levels of algorithmic trading activity, while high trade-to-order volume ratios and average trade size indicate low levels of algorithmic trading. Spread is the difference in the closing ask and closing bid prices divided by the midpoint of the closing ask and closing bid (Chung and Zhang, 2014). Chung and Zhang find that the CRSP-based spread is highly correlated with TAQ-based spread and provides a better approximation than other low-frequency liquidity measures. Volatility is the difference in the daily high price and the daily low price. ETF age is the difference between the ETF's inception year and the year 2022, which is the last year in the sample period.

Table 2 presents descriptive statistics of the main matched sample ETFs. Equity ETFs have an average daily number of trades of 6,683, trade volume of 1.008 million, canceled trades of 540,745, and odd-lot volume of 2,447. Bond ETFs have an average daily number of trades of 1,930, trade volume of 0.45 million, canceled trades of 77,086, and odd-lot volume of 474. On average, hidden volume in bond ETFs is 41% of all trade volume, while hidden volume in equity ETFs is 28%. On average, bond (equity) ETFs have an expense ratio of 0.29% (0.45%), \$4.6 billion (\$4.1 billion) of assets under management and are 9.60 (11.60) years old.

To evaluate the matches, we conduct t-tests on the matching variables. Table 3 panel A presents the results from t-tests on the matching variables for the main sample using daily data.

¹¹ Bartlett and Partnoy (2020) raise the issue with using a ratio as a dependent variable when there may be a case for omitted variable bias and measurement error. As the authors mention, some types of ratios, like percentages, pose little risk of omitted variable bias or measurement error. Our measure of hidden activity are percentages where the denominator of the ratio is total trade volume as provided by MIDAS and unlikely to be measured with error.

There is no difference between the matching variables, which indicates good matches between bond and equity ETFs (p-value of 0.97 for price and 0.95 for market capitalization).

4. Methods

In addition to using the traditional t-test, we use the normalized difference method to test for differences in hidden activity between bond and equity ETFs (Imbens and Wooldridge, 2009). This method is recommended when the sample size is large, which will make even small differences in the covariates statistically significant. We calculate the following normalized difference, Δx , for the hidden activity proxy, *Hidden Volume (%)*, between bond ETFs and equity ETFs,

$$\Delta x = \frac{\bar{X}_B - \bar{X}_E}{\sqrt{S_B^2 + S_E^2}}, \quad (2)$$

where $\bar{X}_B$ and $\bar{X}_E$ are the average of the hidden activity proxy for bond ETFs and equity ETFs, respectively. S_B^2 and S_E^2 are variances of the hidden activity proxy for bond ETFs and equity ETFs, respectively. A normalized difference above 0.25 suggests differences in the covariate distribution between the two groups.

We use a traditional univariate t-test to test for differences in the trading cost metrics between bond and equity ETFs. In a multivariate regression setting, we estimate the following regression:

$$Y_i = \beta_0 + \gamma X_i + \varepsilon_i, \quad (3)$$

where Y_i is the hidden activity proxy, *Hidden Volume (%)* for ETF i . X_i is the set of independent variables including price, market capitalization, total trade volume, spread, volatility, trade size,

cancel-to-trade ratio, trade-to-order-volume ratio, the odd-lot volume ratio, expense ratio, AUM, and fund age (all in logs).

Since the difference in trading dynamics between bond and equity ETFs is a main interest of the paper, we regress the differences in the dependent and independent variables to identify which variables determine differences in trading dynamics between the two groups. Using the same regression specification as above, Y_i is the difference between bond and equity ETFs for the hidden activity proxy (e.g., $Hidden\ Volume\ (\%) = Hidden\ Volume_B\ (\%) - Hidden\ Volume_E\ (\%)$ where B and E represent bond ETFs and equity ETFs, respectively). X_i is the difference in the set of independent variables between the two groups (e.g., $price = price_B - price_E$). We take the absolute value of the control variables before taking the log or the $\log(1+x)$ transformation to use in the difference regressions (due to zero difference values).

5. Empirical Findings

We begin by visually comparing the time-series average of the hidden activity proxy for bond ETFs and equity ETFs. *Hidden Volume ('000)* is obtained for each ETF from Midas for the sample period of January 2021 to March 2022. Figure 1 panel A shows the time-series of average *Hidden Volume ('000)* for bond ETFs and equity ETFs and panel B shows the time-series for *Hidden Volume (%)*. We find that *Hidden Volume ('000)* is of similar levels for bond and equity ETFs. In panel B, *Hidden Volume (%)* is higher in bond ETFs than equity ETFs. *Hidden Volume (%)* in bond ETFs averages around 40%-45% and 25%-30% for equity ETFs over the January 2021 to March 2022 sample period. This indicates that almost half of the trading volume related to bond ETFs is opaque.

5.1 Hidden Activity Differences – Daily

Figure 1 provides visual confirmation for hypothesis 1 in that the percentage of hidden activity appears higher for bond ETFs than equity ETFs. The aggregate hidden activity (hidden volume) in Panel A of figure 1 shows that hidden volume is similar for bond ETFs and equity ETFs, but when overall volume is considered (Panel B), the percentage of hidden volume is higher for bond ETFs than equity ETFs.

Next, we calculate a normalized difference score to test the different level of *Hidden Volume (%)* between bond and equity ETFs. Table 3 panel B reports the normalized difference test on the different levels of hidden activity. Panel B of table 3 reports the average *Hidden Volume (%)* in bond ETFs as 40.8% and in equity ETFs as 27.8%. The absolute value of the normalized difference score is greater than 0.25 for *Hidden Volume (%)* (norm diff = 0.552), which suggests that bond ETFs are traded more opaquely. Table 3 panel C reports the t-test on the difference in the percentage of hidden volume between bond and equity ETFs. On average, bond ETFs have significantly higher percentage hidden volume compared with equity ETFs. These results support hypothesis 1.

5.2 Dark Trading – Daily

Dark pools do not offer pre-trade transparency therefore trading in dark pools can be a substitute for hidden order trading on lit exchanges (Degryse et al., 2021). Figure 2 illustrates the levels of dark trading in bond and equity ETFs. Panel A (dark share quantity) and panel B (number of dark trades) show that dark share quantity is higher in bond ETFs than equity ETFs, but the number of dark trades between the two types of ETFs appear similar.

We test for differences in dark trading between bond and equity ETFs using a t-test. Table 3 panel D presents t-tests on the dark trading variables, dark share quantity and the number of dark trades, between the two groups. Bond ETFs have significantly higher dark share quantity

than equity ETFs (63,422.50 versus 44,895.60, respectively), although the number of dark trades in bond ETFs is smaller than for equity ETF trades (246.7 versus 273.5).¹² This implies that the trade size in dark pools for bond ETFs is higher than that of equity ETFs (as the overall dark quantity is larger for bond ETFs). Consistent with dark trading being a substitute for hidden order trading, dark trading is higher or at similar levels for bond ETFs compared with matched equity ETFs.¹³ The evidence in table 3 supports hypothesis 2 as dark trading activity is higher in bond ETFs than in equity ETFs.

5.3 Other Trading Differences – Daily

To determine possible factors affecting *Hidden Volume (%)* in bond and equity ETFs, we run OLS regressions using *Hidden Volume (%)* as the dependent variable. Table 4 presents OLS regression results on *Hidden Volume (%)* for the main sample with control variables. In table 4, the regression specifications with trade volume, spread, and volatility are presented in columns (1) – (3). The algorithmic trading proxies of trade size, cancel-to-trade ratio, trade-to-order volume ratio, and odd-lot volume ratio are in specifications presented in columns (4) – (6). ETF characteristics including expense ratio, AUM, and fund age are in specifications presented in columns (7) – (9). All independent variables are log-transformed. The difference regressions between bond ETFs and equity ETFs use the absolute value of log-transformed or $\log(1+x)$ transformed differences (due to zero difference values) as independent variables.

The coefficients in column (1) of Table 4 show that an increase in spreads is positively related to *Hidden Volume (%)* in bond ETFs, while volatility is negatively associated with

¹² In the main match sample, equity ETFs have a significantly higher number of dark trades, but results are insignificant in Match 2 and 3 which may indicate that the number of dark trades in bond ETFs and equity ETFs are at similar levels (unreported analyses).

¹³ In unreported tests we find that bond and equity ETFs trade in a similar number of dark pools. Over the sample period, bond ETFs trade on average per week in 15.05 dark pools, while equity ETFs trade on average per week in 15.80 dark pools. Therefore, access to dark trading platforms are not much different between the two types of ETFs.

Hidden Volume (%) in bond ETFs. When volatility is high, bond ETFs are less likely to have hidden trade volume, and when the spread is high, bond ETFs are more likely to have hidden volume. Column (2) shows that equity ETFs with wide spreads and higher trading volume have high *Hidden Volume (%)*, while high volatility is associated with decreases in *Hidden Volume (%)* in equity ETFs. Column (3) in table 4 reports that spread is the main factor affecting the difference in *Hidden Volume (%)* in bond and equity ETFs.

Columns (4) and (5) in table 4 provide mixed evidence regarding algorithmic trading. For bond ETFs, all proxies of AT are negatively related to *Hidden Volume (%)*. In equity ETFs, coefficients for the cancel-to-trade ratio, trade-to-order volume ratio, and odd-lot volume ratio are negative, but the coefficient for trade size is positive. The difference regression estimates shown in column (6) provide conflicting evidence regarding AT and *Hidden Volume (%)*. Overall, we find conflicting evidence regarding the level of AT in bond and equity ETFs in explaining differences *Hidden Volume (%)*.

The coefficients from column (7) in table 4 indicate that younger bond ETFs with low expense ratios and lower AUM have higher *Hidden Volume (%)*. The coefficients from column (8) imply that less mature equity ETFs with high AUM and high expense ratios have high *Hidden Volume (%)*. Column (9) indicates that only AUM and ETF age reflect differences in *Hidden Volume (%)* between bond and equity ETFs. Spread, average trade size, cancel-to-trade ratio, trade-to-order volume ratio, odd-lot volume ratio, AUM, and ETF age influence the differences in *Hidden Volume (%)* between bond and equity ETFs. Overall, these relations provide support for hypothesis 3 in that some trading metrics and fund characteristics relate to differences in hidden trading activity.

5.4 Hidden Activity Differences – Intraday

Next, we use high frequency data to further investigate the factors affecting the differences in *Hidden Volume (%)* between bond and equity ETFs. Table 5 panel A presents descriptive statistics for the intraday sample. The larger sample size for the three-month intraday analysis is due to the less stringent restriction of trading every day for 62 days rather than 314 days in the main sample. Price and market capitalization of the bond ETFs (equity ETFs) are \$48.62 (\$48.42) and 3,994,533 (3,995,931). Table 5 panel B presents the results from the t-test on the matched variables for the intraday sample using high frequency data. The differences between the match variables are not significantly different from zero, which indicate good matches between bond and equity ETFs (p-value of 0.58 for price and 0.99 for market capitalization). Table 5 panel C presents our normalized difference tests. The absolute value of normalized difference score is greater than 0.25 for *Hidden Volume (%)* (norm diff = 0.765), which suggests differences in the levels of *Hidden Volume (%)* between bond and equity ETFs with bond ETFs having higher levels of hidden trading. This is consistent with the main analysis (daily sample from January 2021 through March 2022) which shows that bond ETFs have higher *Hidden Volume (%)* than equity ETFs.

Using the intraday data, we examine trading costs and pinging activity as the intraday proxy for algorithmic trading. In the computation of all spread metrics (computation of the NBBO), we follow Holden and Jacobsen (2014). Quoted spread is the difference in the ask price and the bid price. Percentage quoted spread is the quoted bid-ask spread divided by the midpoint of the quote. Effective spread is the difference between the execution price and the midpoint of the prevailing bid-ask spread. Percentage effective spread is the effective spread divided by the midpoint of the prevailing bid-ask quote. Realized spread is the effective spread minus the change in the quote midpoint five minutes later. The percentage realized spread is the realized

spread divided by the midpoint of the prevailing bid-ask spread. Price impact is the difference between the quote midpoint and the quote midpoint five minutes later. The percentage price impact is the price impact divided by the midpoint of the prevailing bid-ask spread.

5.5 Trading Costs – Intraday

We compare trading costs of bond and equity ETFs. Table 7 provides the t-tests on trading cost metrics between bond and equity ETFs. We find that all trading cost metrics are significantly lower for bond ETFs than equity ETFs. Quoted spread in bond ETFs is lower than in equity ETFs by 0.0296. Effective spread in bond ETFs is lower by 0.0216 and realized spread is lower by 0.0158. Price impact in bond ETFs is lower by 0.0101 compared to equity ETFs. Price impact being smaller for bond ETFs than equity ETFs could be due bond ETF traders having greater hidden activity, as they are avoiding revealing their trading intentions, and the hidden trades leads to a smaller price impact. These statistics imply that bond ETFs are less costly to trade, which may explain the higher proportion of hidden volume in bond ETFs than for equity ETFs.

We examine trading costs in a multivariate regression setting to control for factors such as price, trade size, and volatility. In our regression models, *Hidden Volume (%)* is the dependent variable, and our main variables of interest are the dollar trading cost metrics. Table 7 reports the regression estimations with the control factors (all in logs). The regression specifications testing for the effect of quoted spread, effective spread, realized spread, and price impact on *Hidden Volume (%)* in bond ETFs are in columns (1), (2), (3), and (4), respectively. The regression specifications testing quoted spread, effective spread, realized spread, and price impact on *Hidden Volume (%)* in equity ETFs are in columns (5), (6), (7), and (8), respectively.

Volatility and trade size are negatively related to *Hidden Volume (%)* in bond ETFs. A 1% increase in volatility is associated with a decrease in *Hidden Volume (%)* by 0.05% to 0.08%. *Hidden Volume (%)* in bond ETFs is high when trade size is small and when volatility is low. Quoted spread, effective spread, realized spread, and price impact are all positively related to *Hidden Volume (%)* in bond ETFs. This implies that bond ETFs are traded more opaquely when trading costs are high.

For equity ETFs, volatility and trade size are negatively related to *Hidden Volume (%)* while trade volume is positively related to *Hidden Volume (%)*. Quoted spread, effective spread, realized spread, and price impact are all positively related to *Hidden Volume (%)* in equity ETFs. Overall, our regression analyses indicate that trading cost metrics are positively correlated with the *Hidden Volume (%)* in both types of ETFs.

Next, we are interested in the relation between the differences in trading costs in bond and equity ETFs and if the differences can explain the different levels of *Hidden Volume (%)* between the two types of ETFs. We take the log of the absolute difference in the dependent variable, *Hidden Volume (%)*, and independent variables. The difference regression analyses presented in Table 8 relate the differences in quoted spread, effective spread, realized spread, and price impact with the difference in *Hidden Volume (%)* between bond and equity ETFs.

Interpretation of the absolute value of logged differences is difficult, so we refrain from discussing the magnitude of the coefficients in the difference regression specifications. We see that quoted spread, effective spread, realized spread, and price impact are positively related to *Hidden Volume (%)*. Differences in trading costs can explain some of the different levels of hidden trading in the two types of ETFs. Overall, our results on trading costs support hypothesis

4.

5.6 Pinging Activity – Intraday

We propose that algorithms may try to identify hidden liquidity in our sample ETFs through pinging trades. We extend our analysis using pinging activity as an alternative proxy for AT. We define a ping as a one-share trade. We compare the percentage of one-share trades to total trades (*% of Pings*), average number of pings over the sample period (*Avg # of Pings*), and average number of trades (*Avg # of Trades*) for bond and equity ETFs. Table 9 panel A shows the comparison of pinging activity between the two ETFs. The t-tests indicate that the percentage of one-share trades (*% of Pings*) for equity ETFs (11.89%) is significantly higher than that for bond ETFs (7.41%). The average number of pings for bond ETFs is significantly lower than that of equity ETFs (12,069 versus 45,689, respectively). The average number of trades is 230,034 for bond ETFs and 854,860 for equity ETFs and the difference is statistically significant. Overall, pinging activity is more prevalent in equity ETFs than bond ETFs.

We use the *Avg # of Pings* as the *Pinging* proxy in multivariate regressions, controlling for trade characteristics such as price, market cap, trade volume, and volatility (all variables are in logs). We examine the association between *Pinging* and *Hidden Volume (%)* in bond and equity ETFs. The regression results in table 9 panel B show that *Pinging* is negatively related to *Hidden Volume (%)* for both bond and equity ETFs. The difference in *Pinging* between bond and equity ETFs may explain the different level in *Hidden Volume (%)*. Although high pinging activity in both types of ETFs result in lower hidden trading, the difference in their respective levels of pinging activity can explain why bond ETFs are trading more opaquely. Our results do not support hypothesis 5 and imply that algorithms are not actively seeking to identify hidden orders in ETFs.

6. Robustness

6.1 Out-of-Sample Test

We examine the trading dynamics between bond and equity ETFs in two out-of-sample analyses. The first sample period is from January 2012 to December 2012, and the second sample period is from January 2019 to December 2020. We use the year 2012 as it is the first year the MIDAS data is available, while 2019-2020 encompasses a period before and during the COVID-19 pandemic. The same matching procedure and filters are used for the out-of-sample tests. The price and market capitalization matching specification identifies 40 matched pairs in 2012 and 234 matched pairs in 2019-2020.¹⁴

Figure 3 presents out-of-sample time series graphs of *Hidden Volume (%)* for bond and equity ETFs over the 2012 and 2019-2020 sample periods. *Hidden Volume (%)* is higher in bond ETFs in both out-of-sample periods with spikes during the initial COVID-19 period. The average level of *Hidden Volume (%)* between the bond and equity ETFs during the COVID period is similar to the main sample period, which suggests that volatility is not the main reason we observe a greater percentage of hidden volume for bond ETFs than equity ETFs.

Table 10 panels A and B report normalized difference tests for the 2012 and 2019-2020 sample periods, respectively. Panel A reports the average *Hidden Volume (%)* in bond ETFs is 16.6%. The average *Hidden Volume (%)* in equity ETFs is 10.5%. Normalized difference score of 0.601 indicate that bond ETFs have higher *Hidden Volume (%)* in 2012 than equity ETFs. Panel B reports the average *Hidden Volume (%)* in bond ETFs is 42.95%, which is similar with

¹⁴ The small sample size in 2012 is due to fewer bond ETFs in the market in during that time period, and some of the bond ETFs not passing the filters of trading every day in the sample period and priced above \$5. Bond ETFs had not gained popularity during this time period, and the use of hidden orders were not at the level they are in today's market. Taken together, the 2012 out-of-sample setting should work against us from finding results that support our main conclusion.

our main sample. The average *Hidden Volume (%)* in equity ETFs is 28.87%. Normalized difference score of 0.681 indicates that bond ETFs have higher *Hidden Volume (%)* in 2019-2020.

Our main finding of higher *Hidden Volume (%)* for bond ETFs is consistent over 2012 and 2019-2020. The level of *Hidden Volume (%)* in bond and equity ETFs in the main sample period is higher than the level of *Hidden Volume (%)* in the 2012 sample period, indicating that hidden trading is more prevalent in today's modern markets. Overall, the results in the out-of-sample tests are consistent with the main finding of the paper with bond ETFs having higher *Hidden Volume (%)* relative to equity ETFs.

6.2 Across Exchanges

Lit exchanges in the U.S. compete for order flow with fifteen exchanges competing for market share in the main sample period. The fifteen exchanges in MIDAS include Amex, Arca, Bats-Y, Bats-Z, Boston, CHX, Edge-A, Edge-X, IEX, MEMX, MIAX, NSX, NYSE, Nasdaq, and Phlx. We look across these 15 exchanges to determine if a specific exchange or set of exchanges is driving the high levels of *Hidden Volume (%)* in bond ETFs (Bartlett and O'Hara (2024) document that different exchanges have different levels of hidden activity). Figure 4 illustrates the breakdown of *Hidden Volume (%)* for bond ETFs and equity ETFs across the fifteen exchanges.¹⁵

It appears in figure 4 that the *Hidden Volume (%)* in bond ETFs is reasonably similar to the *Hidden Volume (%)* in equity ETFs for each of the 15 exchanges. In other words, no exchange appears to be catering to hidden trading of bond ETFs, but not equity ETFs, or vice

¹⁵ The *Hidden Volume (%)* for both types of ETFs are first calculated before breaking down the proportion by exchange, so the averages of *Hidden Volume (%)* for the two types of ETFs sum to 1 (or 100%) across exchanges.

versa. Interestingly, Arca has the highest proportion of *Hidden Volume (%)* in both bond and equity ETFs, followed by Nasdaq.

Table 11 reports normalized difference scores for differences in *Hidden Volume (%)* for bond and equity ETFs on each exchange. There is no evidence of higher *Hidden Volume (%)* in bond ETFs compared to equity ETFs on a specific exchange as no normalized difference score is above 0.25. Overall, the high level of *Hidden Volume (%)* in bond ETFs relative to equity ETFs is not driven by a particular exchange.

6.3 By Bond ETF Types

To determine if a particular type of bond ETFs is driving the high level of *Hidden Volume (%)*, we divide the main bond ETF sample into bond ETFs that are considered relatively “more risky” and those considered relatively “less risky”. We consider bond ETFs that track Treasuries, government bonds, and municipal bonds to be less risky and we broadly classified them as Government Bond ETFs. All corporate bond ETFs are included in our riskier Corporate Bond ETF sample. The full sample of equity ETFs in the main sample is used for comparison. The larger sample size in equity ETFs relative to Government Bond ETFs and Corporate Bond ETFs should bias the results against finding significantly higher *Hidden Volume (%)* in the two types of bond ETFs.¹⁶

Table 12 panel A compares *Hidden Volume (%)* in Government Bond ETFs and Corporate Bond ETFs. The average *Hidden Volume (%)* in Corporate Bond ETFs is 41.4%. The average *Hidden Volume (%)* in Government Bond ETFs is 35.2%. The normalized difference score of 0.287 indicates that Corporate Bond ETFs have higher *Hidden Volume (%)* than

¹⁶ The full sample of bond ETFs is not used since some bond ETFs are classified as agency, broad, ABS, etc.

Government Bond ETFs. This result implies that “more risky” bond ETFs may trade more opaquely than “less risky” bond ETFs.

Table 12 panel B shows the comparison of *Hidden Volume (%)* in Equity ETFs and Government Bond ETFs. The average *Hidden Volume (%)* in Equity ETFs is 27.8%, while the average *Hidden Volume (%)* in Government Bond ETFs is 35.2%. Normalized difference score of -0.300 indicates that Government Bond ETFs have higher *Hidden Volume (%)* relative to Equity ETFs. The finding that bond ETFs have higher *Hidden Volume (%)* compared to equity ETFs still holds when comparing a category of relatively “less risky” bond ETFs to equity ETFs.

The comparison of hidden activity in Corporate Bond ETFs and Equity ETFs is presented in panel C of table 12. The average *Hidden Volume (%)* in Equity ETFs is 27.8%, while the average *Hidden Volume (%)* in Corporate Bond ETFs is 41.4%. Normalized difference score of -0.583 indicates that Corporate Bond ETFs have higher *Hidden Volume (%)* than equity ETFs. Overall, bond ETFs have higher hidden activity compared to equity ETFs, regardless of whether the underlying bonds are relatively more or less risky.

To summarize our robustness tests, the 2012 out-of-sample analysis shows that bond ETFs have higher *Hidden Volume (%)* than equity ETFs during a time when there were few bond ETFs in existence and when hidden trading was not as high as in today’s markets. Using a more recent out-of-sample period with an exogenous volatility shock (COVID-19), we once again find that bond ETFs are traded more opaquely compared to equity ETFs. *Hidden Volume (%)* are around the same level for equity ETFs and bond ETFs across exchanges. We find some evidence that “riskier” bond ETFs have higher *Hidden Volume (%)* but even “less risky” bond ETFs have higher *Hidden Volume (%)* than equity ETFs.¹⁷

¹⁷ In an unreported analysis, we conduct an event study of hidden trading activity in bond ETFs and equity ETFs around Federal Open Market Committee (FOMC) announcements in the main sample period (only one meeting

7. Conclusion

ETFs have gained interest in the literature (Ben-David, Franzoni, and Moussawi; 2018; Box, Davis, Evans, and Lynch, 2021; Huang, O'Hara and Zhong, 2021; Karmaziene and Sokolovshi, 2022; Li and Zhu, 2022). This research on ETFs primarily focuses on equity ETFs with few studies examining bond ETFs (Dannhauser, 2017; Converse, Levy-Yeyati, and Williams, 2023). In this study, we are interested in comparing the trading dynamics of bond ETFs and equity ETFs. We find a puzzle: *Hidden Volume (%)* is greater in bond ETFs than for comparable equity ETFs.

We suspect different trading patterns among bond and equity ETFs, because the underlying securities trade in different markets and are fundamentally different asset classes. Bonds trade primarily in the OTC market structure with no pre-trade transparency and bond markets are relatively less liquidity than equity markets. Equities trade on liquid, lit exchanges where price quotes are publicly available with the option to hide orders. Equities can also be traded off-exchange in dark pools where price quotes are not publicly posted. ETFs trade on the same platforms as equities. Therefore, when a type of security (bonds) has access to another market with different features (pre-trade price transparency and hidden limit orders), it is of interest how it trades in that new market.

We examine the level of hidden activity in bond ETFs and equity ETFs. Empirical evidence confirms differences in trading dynamics between bond ETFs and equity ETFs. Specifically, bond ETFs have a larger *Hidden Volume (%)* than comparable equity ETFs.

resulted in a rate change during the period). The level of *Hidden Volume (%)* for bond ETFs and equity ETFs during the [-5, +5] event window around the announcements is around the same level as in the main sample period (40% for bond ETFs and 28% for equity ETFs).

Consistent with dark trading in off-exchange venues being a substitute for hidden trading on exchanges, bond ETFs have higher levels of dark volume than equity ETFs. Some factors that may explain the difference in *Hidden Volume (%)* between bond and equity of ETFs include spread, trade size, cancel-to-trade ratio, trade-to-order volume ratio, odd-lot volume ratio, AUM, and ETF age.

We find that bond ETFs have lower trading costs than equity ETFs, and the differences in trading costs do explain some of the different levels *Hidden Volume (%)* between bond and equity ETFs. We find evidence that pinging activity (one share trades) can also explain some of the higher *Hidden Volume (%)* in bond ETFs than equity ETFs although pinging activity is negatively related to *Hidden Volume (%)* in both types of ETFs. The finding of higher *Hidden Volume (%)* in bond ETFs relative to equity ETFs is persistent throughout the sample period and in two out-of-sample analyses. Results are not driven by a specific exchange nor by certain types of bond ETFs.

In conclusion, it is a puzzle as to why bond ETFs are traded more opaquely than equity ETFs. Bonds are harder to value due to infrequent trading in the OTC bond market, which may affect how bond ETFs trade. Alternatively, traders of bond ETFs may simply have a preference for trading bonds (hence, bond ETFs) opaquely. We conclude that although bond ETFs are a more transparent trading option than OTC bonds, traders do not appear to be taking full advantage of this transparency. Furthermore, high levels of hidden trading activity in bond ETFs may provide some additional insight into the slow adoption of electronic trading in bonds (O'Hara and Zhou, 2021).

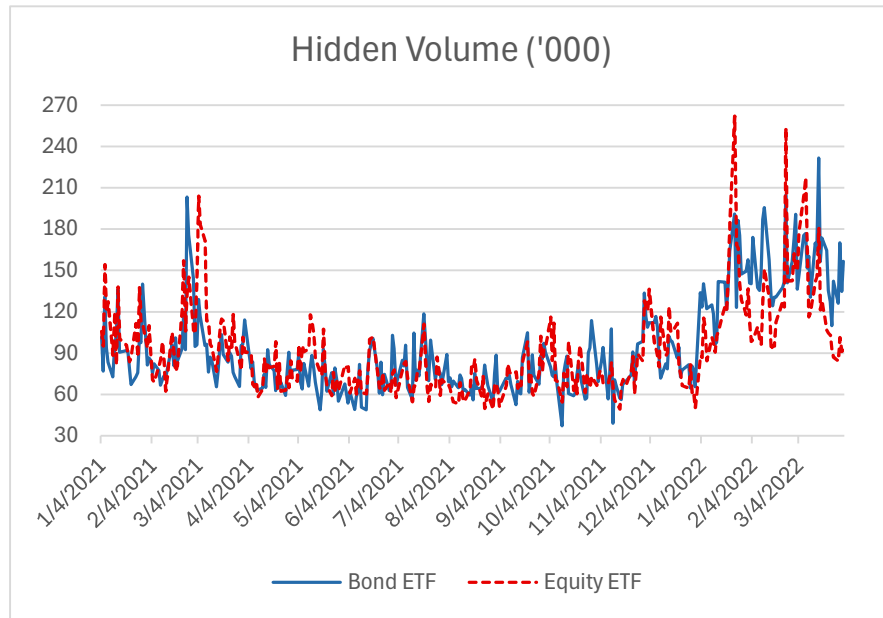
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Panel A. Hidden Volume ('000)



Panel B. Hidden Volume (%)

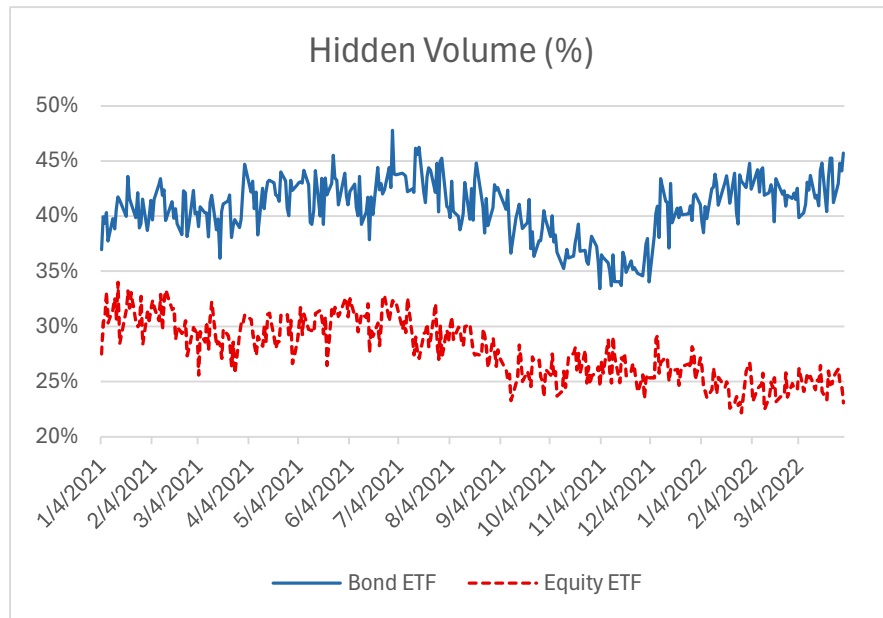
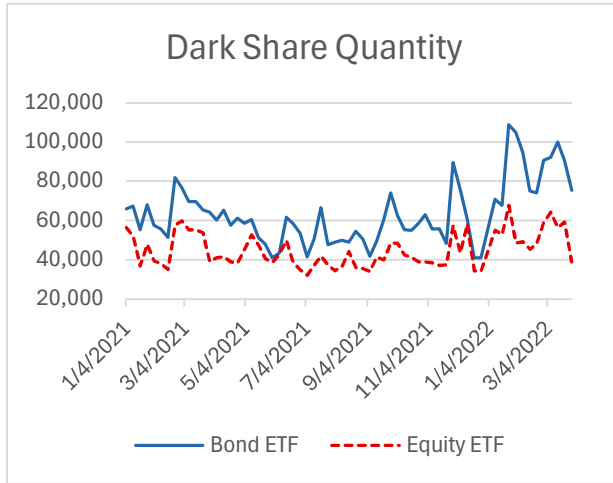


Figure 1. Time-series of Hidden Activity in Bond ETFs and Equity ETFs.

This figure shows the time-series of hidden activity between bond ETFs and equity ETFs in the sample period from January 2021 to March 2022. Panel A graphs total hidden volume (in thousands of shares) and panel B graphs hidden volume as a percentage of total volume. The solid blue (dashed red) line denotes hidden volume in bond ETFs (equity ETFs).

Panel A. Dark Share Quantity



Panel B. Number of Dark Trades

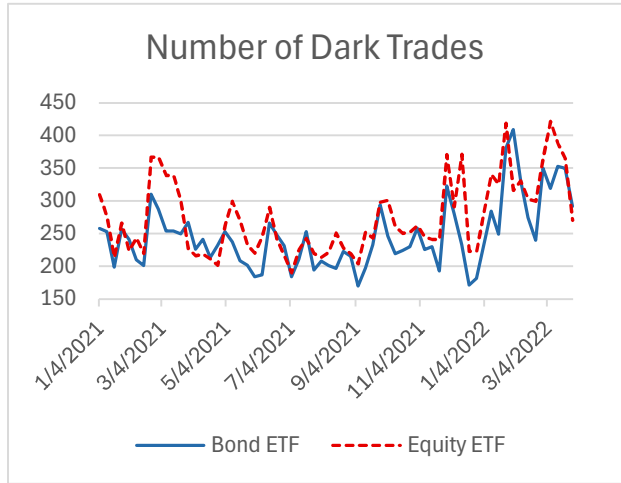
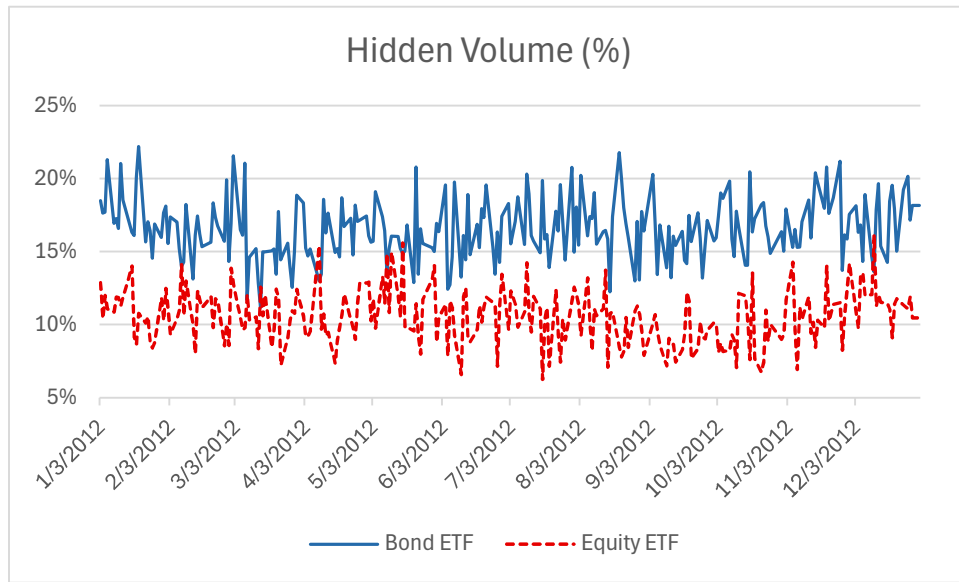


Figure 2. Time-series of Dark Trading Activity in Bond ETFs and Equity ETFs.

This figure shows the time-series of dark trading activity between bond ETFs and equity ETFs in the sample period from January 2021 to March 2022. In panel A, the solid blue (dashed red) line denotes dark share quantity for bond ETFs (equity ETFs). In panel B, the solid blue (dashed red) line denotes the number of dark trades in bond ETFs (equity ETFs).

Panel A. Hidden Volume (%) - 2012



Panel B. Hidden Volume (%) – 2019-2020

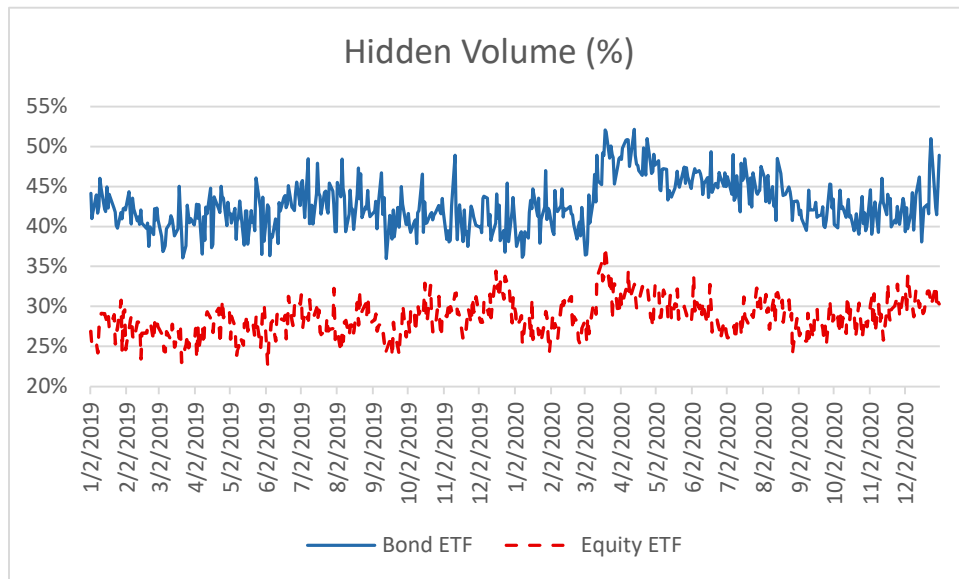


Figure 3. Time-series of Trading Dynamics in Bond ETFs and Equity ETFs Out of Sample.

This figure shows the time-series of various trading dynamics between bond ETFs and equity ETFs out of sample. Panel A graph the sample period from January 2012 to December 2012 and panel B graph the sample period from January 2019 to December 2020. The solid blue (dashed red) line denotes hidden volume as a percentage of total volume for the sample of bond ETFs (equity ETFs).

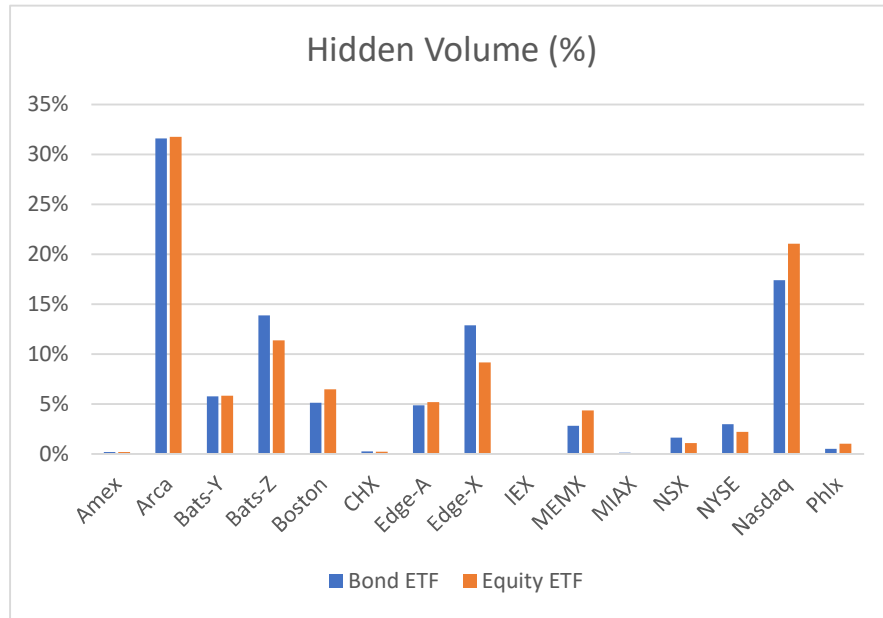


Figure 4. Hidden Trading Activity in Bond ETFs and Equity ETFs Across Exchanges.

This figure shows the hidden trading activity between bond ETFs and equity ETFs across exchanges in the sample period from January 2021 to March 2022. The blue (orange) bar denotes Hidden Volume (%) for bond ETFs (equity ETFs).

Table 1: Types of Bond ETFs

This table reports the types of bond ETFs in the full sample. Panel A reports the counts for the types of bonds held in the bond ETFs. The classification for bond type is found on the ETF Database under FactSet Classifications for Category. Panel B reports the counts for the grade of the bonds held in the bond ETFs. The classification for bond grade is found on the ETF Database under FactSet Classifications for Focus.

Panel A: Type	Count
ABS	3
Agency	1
Broad	50
Convertible	2
Corporate	74
Government	20
Municipal	16
Treasury	28
Total	194
Panel B: Grade	
Broad	28
High-Yield	36
Investment	130
Total	194

Table 2: Descriptive Statistics

This table reports descriptive statistics of bond and equity ETFs over the daily sample period from January 2021 to March 2022. Price and market capitalization are from CRSP. Spread is the difference in the closing ask and closing bid prices divided by the midpoint of the closing ask and closing bid from CRSP. Volatility is the difference in the daily high price and the daily low price from CRSP. Number of trades, trade size, trade volume ('000), number of canceled orders, and odd-lot trades are sourced from Midas. Hidden Volume (%) is the proportion of hidden volume in bond or equity ETFs over total volume in bond or equity ETFs. Lit Volume (%) is the proportion of lit volume in bond or equity ETFs over total volume in bond or equity ETFs. Expense ratio, asset under management (AUM), and ETF inception year are from ETF Database. Fund age is the difference between the ETF's inception year and the year 2022, which is the last year in the sample period.

Variable	Bond ETFs			Equity ETFs		
	Mean	Median	Std. Dev	Mean	Median	Std. Dev
Price	52.78	50.52	29.58	52.79	49.09	29.68
Market Cap ('000)	3,970,471	1,050,052	7,469,574	3,967,613	1,062,371	7,522,108
Spread	0.001	0.001	0.001	0.001	0.001	0.001
Volatility	0.22	0.13	0.29	0.82	0.58	0.77
Number of Trades	1,930.33	261.77	6,904.23	6,682.77	511.58	24,654.98
Trade Size	184.37	142.81	119.13	141.30	117.43	85.58
Trade Volume ('000)	449.97	45.48	52.42	1,008.92	52.42	3,257.77
Canceled Orders	77,086.08	10,163.88	241,964.51	540,745.47	191,202.05	1,360,227.90
Odd-Lots	473.70	54.94	1,789.66	2,447.20	261.28	10,250.23
Hidden Volume (%)	0.41	0.41	0.15	0.28	0.23	0.19
Lit Volume (%)	0.59	0.60	0.15	0.72	0.77	0.18
Expense Ratio (%)	0.29	0.20	0.28	0.45	0.47	0.26
AUM (in millions)	4,586.56	1,278.85	8,307.80	4,102.04	980.65	8,301.20
Fund Age	9.60	9.50	4.17	11.60	11.00	6.61
Observations	194			194		

Table 3: Significance Test of Differences - Daily

This table reports the test of differences in the match variables, hidden activity, and dark trading between bond and equity ETFs. Panel A presents the t-test on the match variables. Panel B presents the normalized difference test of Hidden Volume (%). A normalized difference above 0.25 in absolute value suggests differences in the covariate distribution between the two groups. Panel C reports the t-test on the differences in Hidden Volume (%). Panel D reports the t-test on the differences in dark trading variables. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Differences in Match Variables					
	Equity ETF	Bond ETF	Difference	t-value	p-value
Price	52.79	52.78	0.01	0.03	0.97
Market Cap ('000)	3,967,613	3,970,471	-2,858	-0.07	0.95
Observations	194	194			
Panel B: Normalized Difference Test of Hidden Activity					
	Equity ETFs		Bond ETFs		Norm Diff
	Mean	Std.	Mean	Std.	
Hidden Volume (%)	27.8	18.5	40.8	14.6	-0.552
Observations	194		194		
Panel C: T-test of Hidden Activity					
	Equity ETFs	Bond ETFs	Difference	t-value	
Hidden Volume (%)	27.8	40.8	-12.9***	-93.62	
Observations	60,916	60,916			
Panel D: Differences in Dark Trading					
	Equity ETFs	Bond ETFs	Difference	t-value	
Dark Share Quantity	44,895.6	63,422.5	-18,526.9***	-8.10	
# of Dark Trades	273.5	246.7	26.74***	2.75	
Observations	194		194		

Table 4: Hidden Volume (%) OLS Regression – Daily

The table reports the OLS regression analyses to examine the determinants for the differences in Hidden Volume (%) between bond and equity ETFs. Variables include the ETFs' price, market capitalization ('000), total trade volume, spread, volatility, trade size, cancel-to-trade ratio, trade-to-order-volume ratio, the odd-lot volume ratio, expense ratio, AUM, and ETF age (all in logs). The difference specifications use the absolute value of the variables before taking the log or log(1+x) transformation. T-statistics are recorded in parentheses and asterisks ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

VARIABLES	Hidden Volume (%)								
	Bond ETF (1)	Equity ETF (2)	Difference (3)	Bond ETF (4)	Equity ETF (5)	Difference (6)	Bond ETF (7)	Equity ETF (8)	Difference (9)
Price	-0.1420*** (-7.0058)	0.1070*** (18.6002)	-0.0008 (-0.8438)	-0.1020*** (-5.5558)	0.1221*** (23.4875)	-0.0007 (-0.7341)	0.0394*** (19.2359)	0.0184*** (9.6047)	0.0094*** (9.0677)
Market Cap	-0.0045 (-1.1118)	-0.0786*** (-27.3294)	0.0061*** (6.3498)	-0.0261*** (-7.0916)	-0.0711*** (-27.4405)	0.0072*** (7.6423)	0.0081*** (5.3770)	-0.0958*** (-56.0617)	0.0169*** (20.2054)
Trade Volume	-0.0011 (-1.1775)	0.0117*** (13.4664)	-0.0010 (-1.1225)						
Spread	0.0259*** (17.3682)	0.0239*** (18.4213)	12.1611*** (14.7659)						
Volatility	-0.0177*** (-10.7320)	-0.0268*** (-19.1809)	-0.0018 (-1.0570)						
Trade Size				-0.0167*** (-6.2129)	0.0860*** (35.9511)	-0.0026*** (-2.7804)			
Cancel-to-Trade				-0.0054*** (-3.4982)	-0.0748*** (-48.6347)	-0.0106*** (-11.5588)			
Trade-to-Order Vol				-0.0709*** (-59.3977)	-0.1170*** (-94.4549)	-0.0368*** (-47.6078)			
Odd-Lot Vol				-0.0489*** (-38.6907)	-0.0299*** (-22.9901)	0.0020** (2.2230)			
Expense Ratio							-0.0205*** (-18.2586)	0.0060*** (4.3311)	0.0040 (0.5421)
AUM							-0.0381*** (-25.5066)	0.0438*** (26.2053)	0.0151*** (18.4622)
ETF Age							-0.0739*** (-35.0835)	-0.0692*** (-46.5756)	0.0141*** (8.2132)
Constant	1.1754*** (12.2923)	1.0678*** (28.9161)	0.0701*** (6.0867)	1.2534*** (14.1128)	0.5513*** (14.9496)	0.0611*** (4.7835)	0.5412*** (43.9940)	1.3998*** (108.4971)	-0.1930*** (-21.8268)
Observations	60,904	60,865	60,876	60,583	60,765	60,877	60,916	60,916	60,877
R-squared	0.0072	0.0226	0.0042	0.1534	0.1769	0.0421	0.0668	0.2191	0.0335
ETF FE	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
F-Stat	88.14	280.10	51.74	1,824	2,169	444.80	872.10	3,417	422.20

Table 5: Descriptive Statistics and Tests of Differences - Intraday

This table reports descriptive statistics and tests of differences in the match variables and hidden activity between bond and equity ETFs. The intraday sample period is from January 2022 to March 2022. Panel A reports summary statistics. Panel B presents the t-test on the match variables. Panel C presents the normalized difference test of Hidden Volume (%) between bond and equity ETFs. A normalized difference above 0.25 in absolute value suggests differences in the covariate distribution between the two groups. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Descriptive Statistics						
	Bond ETFs			Equity ETFs		
Variable	Mean	Median	Std. Dev	Mean	Median	Std. Dev
Price	48.262	47.471	27.512	48.422	45.482	26.89
Market Cap (‘000)	3,994,533.5	724,614.52	9,858,604.3	3,995,931	701,685.22	9,999,135.9
Number of Trades	2,187.274	213.984	8571.11	10,395.674	378.145	43,064.347
Trade Size	178.106	137.858	132.265	136.976	107.537	99.851
Trade Volume (‘000)	531.521	30.615	1,875.194	1,623.752	46.611	6,152.361
Canceled Orders	82,713.966	9,336.161	269,918.64	681,689.76	229,764.34	1,484,643.1
Odd-Lots	530.2	52.694	2,350.596	3,933.722	164.355	17,821.356
Hidden Volume (%)	43.5	43.7	16.1	25.2	21.3	17.7
Lit Volume (%)	56.5	56.3	16.1	74.8	78.7	17.7
Observations	293			293		
Panel B: Differences in Match Variables						
	Equity ETF	Bond ETF	Difference	t-value	p-value	
Price	48.42	48.26	0.16	0.55	0.58	
Market Cap (‘000)	3,995,931	3,994,533	1,398	0.01	0.99	
Observations	293	293				
Panel C: Normalized Difference Test of Hidden Activity						
	Equity ETFs		Bond ETFs		Norm Diff	
	Mean	Std.	Mean	Std.		
Hidden Volume (%)	25.2	17.7	43.5	16.1	-0.765	
Observations	293		293			

Table 6: Trading Costs – Intraday

This table reports the t-test of the difference in various trading costs metrics between bond and equity ETFs. Quoted spread is the quoted bid-ask spread associated with the transaction. Percentage quoted spread is the quoted bid-ask spread divided by the midpoint of the quote. Effective spread is the difference between the execution price and the midpoint of the prevailing bid-ask spread. Percentage effective spread is the effective spread divided by the midpoint of the prevailing bid-ask quote. Realized spread is the effective spread minus the change in the quote midpoint five minutes later. The percentage realized spread is the realized spread divided by the midpoint of the prevailing bid-ask spread. Price impact is the difference between the quote midpoint and the quote midpoint five minutes later. The percentage price impact is the price impact divided by the midpoint of the prevailing bid-ask spread. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Variable	Bond ETF	Equity ETF	Difference	t-stat
Quoted Spread	0.0430	0.0725	-0.0296	7.39***
% Quoted Spread	0.0010	0.0018	-0.0008	7.23***
Effective Spread	0.0239	0.0455	-0.0216	7.42***
% Effective Spread	0.0006	0.0011	-0.0006	7.65***
Realized Spread	0.0192	0.0350	-0.0158	5.75***
% Realized Spread	0.0005	0.0009	-0.0004	6.17***
Price Impact	0.0163	0.0265	-0.0101	4.30***
% Price Impact	0.0004	0.0007	-0.0003	4.89***
Observations	293	293	293	

Table 7: Hidden Volume OLS Regression - Intraday

The table reports the OLS regression analyses used to examine the determinants for the level of Hidden Volume (%) between bond and equity ETFs over the sample period of January 2022 to March 2022. Trade characteristics include the ETFs' price, market capitalization, total trade volume, volatility, trade size (all in logs). Metrics of dollar trading costs include quoted spread, effective spread, realized spread, and price impact (all in logs). T-statistics are recorded in parentheses and asterisks ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Bond ETFs				Equity ETFs			
	Hidden Volume (%)				Hidden Volume (%)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Intercept	0.8967*** (8.09)	1.1189*** (9.96)	1.0951*** (9.70)	1.0302*** (10.10)	0.3439*** (2.74)	0.7248*** (5.38)	0.7181*** (4.84)	0.3204** (2.32)
Price	-0.0570*** (-2.80)	-0.0543** (-2.47)	-0.0542** (-2.48)	-0.0457** (-2.30)	-0.0516** (-2.08)	-0.0835*** (-2.97)	-0.0664** (-2.19)	0.0006 (0.02)
Market Cap	0.0353*** (2.99)	0.0413*** (3.28)	0.0393*** (3.14)	0.0268** (2.31)	0.0220*** (2.84)	0.0323*** (3.60)	0.0250*** (2.60)	0.0001 (0.02)
Trade Volume	0.0003 (0.02)	-0.0247 (1.57)	-0.0240 (1.54)	0.0020 (0.14)	0.0354*** (4.24)	0.0243** (2.57)	0.0109 (1.09)	0.0410*** (4.34)
Volatility	-0.0838*** (-6.40)	-0.0625*** (-4.29)	-0.0484*** (-3.88)	-0.0650*** (-5.69)	-0.1091*** (-5.84)	-0.0971*** (-4.58)	-0.0666*** (-2.97)	-0.1112*** (5.48)
Trade Size	-0.0836*** (-3.48)	-0.0845*** (-3.29)	-0.0795*** (-3.11)	-0.0933*** (-3.95)	-0.0345* (-1.68)	-0.0604** (-2.60)	-0.0462* (-1.84)	-0.0503** (2.28)
Quoted Spread	0.1317*** (7.61)				0.2052*** (16.45)			
Effective Spread		0.0795*** (4.00)				0.2042*** (12.23)		
Realized Spread			0.0643*** (4.28)				0.1314*** (9.50)	
Price Impact				0.1038*** (8.51)				0.1238*** (14.03)
Observations	293	293	293	293	293	293	293	293
Adj R-Squared	0.3862	0.3006	0.3059	0.4113	0.6325	0.5281	0.4515	0.5752
F-Stat	31.41	21.77	22.30	34.77	81.89	53.59	39.69	64.65

Table 8: Regressions using Differences - Intraday

This table reports OLS regressions of the differences in Hidden Volume (%) between bond and equity ETFs over the sample period of January 2022 to March 2022. The dependent variable of interest is the difference in hidden volume between bond and equity ETFs. Trade characteristics include the differences in ETFs' price, market capitalization, total trade volume, volatility, trade size. Metrics of dollar trading costs include the differences in quoted spread, effective spread, realized spread, and price impact. All difference specifications use the log of the differences between bond and equity ETFs. T-statistics are recorded in parentheses and asterisks ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Hidden Volume (%) (Bond ETF – Equity ETF)			
	(1)	(2)	(3)	(4)
Intercept	0.3687*** (5.81)	0.2814*** (4.06)	0.2462*** (3.60)	0.2342*** (3.76)
Price	-0.0011 (-0.13)	0.0039 (0.42)	0.0033 (0.35)	-0.0100 (-1.13)
Market Cap	-0.0053 (-0.87)	-0.0085 (-1.27)	-0.0083 (-1.23)	-0.0045 (-0.71)
Trade Volume	-0.0010 (-0.17)	0.0073 (1.11)	0.0095 (1.45)	0.0086 (1.41)
Volatility	0.0326*** (2.87)	0.0258** (2.09)	0.0223* (1.80)	0.0247** (2.10)
Trade Size	-0.0151* (1.92)	-0.0173** (-2.02)	-0.0180** (-2.08)	-0.0181** (-2.21)
Quoted Spread	1.3062*** (8.72)			
Effective Spread		0.8566*** (4.41)		
Realized Spread			0.7366*** (3.68)	
Price Impact				2.0368*** (6.89)
Observations	293	293	293	293
Adj R-Squared	0.2261	0.0823	0.0640	0.1596
F-Stat	15.17	5.35	4.32	10.21

Table 9: Pinging Activity

This table reports the intraday results of algorithmic trading using pinging activity as the proxy. Panel A of this table reports t-tests of the differences in pinging activity between bond and equity ETFs over our sample period of January 2022 to March 2022. % of Pings is the proportion of the number of one share trades to total trades per day and Avg # of Pings is the average of pings over the sample period. Panel B reports the regressions of pinging activity on Hidden Volume (%) for bond and equity ETFs over our sample period. Trade characteristics include the ETFs' price, market capitalization, total trade volume, volatility, trade size (all in logs). Pinging is defined as the average number of pings over the sample period. The difference regression specification uses the log of the absolute difference in the dependent variable and independent variables between bond and equity ETFs. T-statistics are reported in parentheses and asterisks ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Bond ETF	Equity ETF	Difference	t-stat
Panel A				
% of Pings	7.41%	11.89%	-4.48%	7.95***
Avg # of Pings	12,069.10	45,688.59	-33,619.49	3.19**
Avg # of Trades	230,034.10	854,860.13	-624,826.04	3.11**
Hidden Volume (%)				
	Bond ETF	Equity ETF	Difference	
	(1)	(2)	(3)	
Panel B				
Intercept	1.0451*** (10.69)	1.4070*** (8.81)	0.1442** (2.17)	
Price	-0.0027 (-0.13)	-0.0657** (-2.06)	0.0008 (0.08)	
Market Cap	0.0418*** (3.89)	0.0081 (0.86)	-0.0074 (-1.14)	
Number of trades	-0.0165 (-1.00)	0.0004 (0.02)	0.0009 (0.11)	
Volatility	-0.0487 (-1.37)	0.0054 (0.22)	0.0098 (0.74)	
Trade Size	-0.1168*** (-5.81)	-0.1185*** (-4.23)	-0.0175** (-2.03)	
Pinging	-0.0504*** (-3.97)	-0.0513*** (-3.21)	0.0209** (2.13)	
Observations	293	293	293	
Adj R-Squared	0.3355	0.3235	0.0327	
F-Stat	25.40	23.47	2.64	

Table 10: Out-of-Sample Robustness Tests

This table reports the normalized difference test on hidden activity for two out-of-sample tests. Panel A presents the results using the January 2012 to December 2012 sample period, and panel B presents the results using the January 2019 to December 2020 sample period. A normalized difference above 0.25 in absolute value suggests differences in the covariate distribution between the two groups.

	Bond ETFs		Equity ETFs		Normalized Difference
Variable	Mean	Std. Dev.	Mean	Std. Dev.	
Panel A: Sample Period 2012					
Hidden Volume (%)	16.6	7.7	10.5	6.6	0.601
Observations	40		40		
Panel B: Sample Period 2019-2020					
Hidden Volume (%)	42.95	11.84	28.87	16.94	0.681
Observations	234		234		

Table 11: Differences in Hidden Activity across Exchanges

This table reports the normalized difference test on hidden activity between bond and equity ETFs across lit exchanges in the sample period of January 2021 to March 2022. Panel A reports the level of Hidden Volume (%) on each exchange. A normalized difference above 0.25 in absolute value suggests differences in the covariate distribution between the two groups.

	Bond ETFs		Equity ETFs		Normalized Difference
	Mean	Std. Dev.	Mean	Std. Dev.	
Hidden Volume (%)					
Amex	0.19	0.80	0.18	0.71	0.0036
Arca	31.59	21.45	31.75	23.48	-0.0050
Bats-Y	5.75	8.27	5.82	9.22	-0.0053
Bats-Z	13.86	13.98	11.38	12.69	0.1317
Boston	5.14	8.01	6.49	10.26	-0.1039
CHX	0.26	3.03	0.24	2.71	0.0050
Edge-A	4.89	6.99	5.18	7.40	-0.0291
Edge-X	12.87	14.27	9.18	10.99	0.2049
IEX	0.00	0.00	0.00	0.00	0.0000
MEMX	2.84	6.89	4.36	10.02	-0.1254
MIAX	0.12	1.16	0.02	0.26	0.0845
NSX	1.62	4.65	1.10	3.36	0.0919
NYSE	2.98	5.33	2.22	4.71	0.1065
Nasdaq	17.39	14.89	21.07	17.93	-0.1579
Phlx	0.52	2.03	1.02	2.97	-0.1410
Total	100%		100%		

Table 12: Differences in Hidden Activity by ETF Types

This table reports the normalized difference test on hidden activity between different bond ETF types and equity ETFs in the sample period of January 2021 to March 2022. Corporate Bond ETFs consist of all ETFs that hold corporate bonds and Government Bond ETFs consist of all ETFs that are classified as government ETFs, Treasury ETFs, and Municipal ETFs. Equity ETFs consist of the full sample of equity ETFs. Panel A compares Hidden Volume (%) in Corporate Bond ETFs and Government Bond ETFs. Panel B compares Hidden Volume (%) in Equity ETFs and Government Bond ETFs. Panel C compares Hidden Volume (%) in Equity ETFs and Corporate Bond ETFs. A normalized difference above 0.25 in absolute value suggests differences in the covariate distribution between the two groups.

Panel A:					
	Corporate Bond ETFs		Government Bond ETFs		Normalized Difference
Variable	Mean	Std. Dev.	Mean	Std. Dev.	
Hidden Volume (%)	41.4	14.2	35.2	16.3	0.287
Observations	80		76		
Panel B:					
	Equity ETFs		Government Bond ETFs		Normalized Difference
Variable	Mean	Std. Dev.	Mean	Std. Dev.	
Hidden Volume (%)	27.8	18.5	35.2	16.3	-0.300
Observations	194		76		
Panel C:					
	Equity ETFs		Corporate Bond ETFs		Normalized Difference
Variable	Mean	Std. Dev.	Mean	Std. Dev.	
Hidden Volume (%)	27.8	18.5	41.4	14.2	-0.583
Observations	194		80		